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Perspective

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From air quality forecasting to integrated and intelligent urban prediction systems

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Air quality forecasting has evolved into a multi-component enterprise integrating chemical transport models, emissions, observations, data assimilation, and artificial intelligence. We argue that future progress will depend less on refinement of individual components than on integrating their interfaces across urban physics, atmospheric chemistry, machine learning, dense observation networks, human activity, and decision-making systems. We outline ten emerging directions that together constitute a transition towards integrated and intelligent urban prediction systems.

Air pollution remains one of the most consequential environmental risks to human health, contributing to several million premature deaths annually and imposing substantial economic and environmental costs worldwide[1,2]. Reliable air quality forecasts have become a cornerstone of public health and environmental protection, providing early warning of pollution episodes, guiding mitigation strategies, and supporting policy decisions at local, national and global scales.

Atmospheric composition in cities evolves through nonlinear interactions among emissions, photochemistry, deposition, transport, turbulence, and other meteorological forcings (e.g., radiation, temperature, humidity), all of which are strongly modulated by urban, industrial, and other human activities. Over the past few decades, advances in scientific understanding, numerical models, observing systems, and computing power have transformed air quality prediction from a largely empirical practice into a model-based scientific and operational framework[3–5]. Modern operational systems combine chemical transport models, detailed and increasingly dynamic emission inventories, in-situ and space-based observations, meteorological analyses, data assimilation, and, more recently, artificial intelligence (AI) components within unified prediction frameworks.

This evolution has resulted in the conventional air quality prediction system illustrated in Fig. 1. Underlying this development has been an implicit assumption: that more detailed representation of processes, finer spatial resolution, and larger amounts of data would lead to greater predictive skill[6–8]. Recent experience suggests that this assumption is now reaching its limits. The marginal returns from increasing chemical mechanism details, refining grid spacing below the resolution of the input emission data, or adding generic data-driven components are no longer commensurate with the resources invested. Although some modelling systems already use data assimilation, ensemble forecasting, dynamic emission adjustment, near-real-time observations, probabilistic post-processing, coupled chemistry-meteorology and exposure related applications, in most cases these individual components are not integrated all together into operational systems. Future advances will come from a fundamental rethinking of our modelling approach. Further, many persistent discrepancies in current forecasts cannot be attributed solely to deficiencies in individual components, but also arise from inconsistencies at the interfaces between components. Another subject of concern is the inconsistency between calculated concentrations and actual human exposure.

Here, we argue that the next phase of progress in urban air quality forecasting will be defined by the integration of components rather than by their independent elaboration and improvement. The transition is from forecasting systems composed of separately optimized parts towards what we call integrated and intelligent urban prediction systems (digital twins): frameworks in which urban computational fluid dynamics (CFD), machine learning, observational networks, human activity patterns, and decision interfaces are co-designed. We develop this argument in four steps: first tracing the historical evolution of air quality prediction and the maturation of its component technologies, then identifying the principal challenges currently limiting forecast capability and outlining ten emerging directions through which integration at the interfaces is beginning to take shape, and finally summarizing what an integrated and intelligent paradigm would mean for the practice and purpose of urban air quality prediction.

Historical Evolution and Model Components

The architecture of modern air quality forecasting reflects a layered accumulation of scientific advances, each of which introduced a new system component while inheriting the limitations of those that came before. Understanding the current state of the field, and the limits of incremental refinement, requires a brief account of how the principal components emerged and how they have been integrated.

The Empirical Approaches

The earliest systematic concerns with urban air pollution were driven by acute episodes associated with industrialization in Europe and North America (e.g., the 1930 Meuse Valley pollution episode in Belgium[9], the

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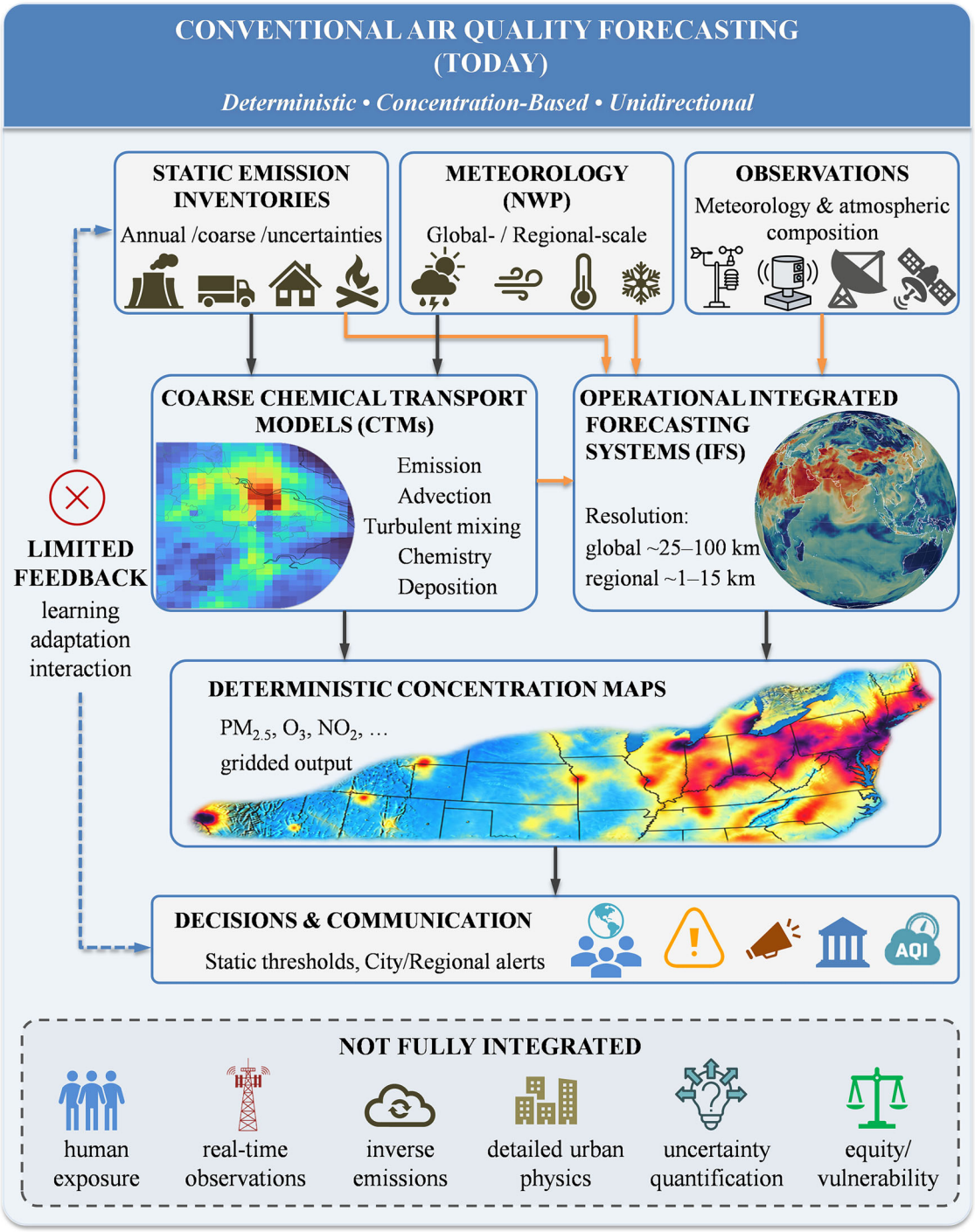


Fig. 1 | Conventional air quality forecasting system. The schematic summarizes the dominant architecture of current air quality forecasting systems, in which emission inventories and meteorological fields are ingested into chemical transport models (CTMs) to simulate concentration fields of atmospheric chemical species, while the Integrated Forecasting System (IFS) performs data assimilation using observations. The generated operational forecasts of atmospheric composition are used for public communication and decision-making. Feedback, adaptive learning and interaction between forecasts and decisions remain limited, while capabilities such as human-exposure assessment, near-real-time observational constraints, inverse emission estimation, detailed urban physics, uncertainty quantification and equity or vulnerability information are not yet fully integrated across conventional operational systems.

1948 Donora smog in the United States[10], and the 1952 Great Smog of London, UK[11]). Prediction of events that were linked to stagnant boundary layer conditions combined with concentrated emissions from industrial and residential combustion, was largely empirical and based on simple statistical regression models[12,13]. This empirical approach, which linked pollutant concentrations to meteorological predictors (i.e., mixing height, wind speed, temperature) provided useful short-term warnings, but lacked physical generality and realism[14]. Therefore, such approaches could not be extended to new chemical regimes or unfamiliar urban contexts.

Chemical Transport Models

The advent of numerical weather prediction in the 1960s and 1970s laid the mathematical and computational foundation for physics-based modelling. As computational resources expanded and new information on chemical

mechanisms became available, atmospheric scientists coupled transport equations with simplified chemical mechanisms, creating the first chemical transport models (CTMs)[15,16]. These models explicitly represented emissions, advection, turbulent mixing, chemical transformation, and deposition as coupled processes. In the 1980s and 1990s, regional CTMs such as the Community Multiscale Air Quality (CMAQ[17]) model in the United States and the European Monitoring and Evaluation Program (EMEP[18]) model in Europe enabled systematic analysis of ozone, particulate matter, and acid deposition. Originally developed for policy assessment rather than real-time forecasting, the use of these models has since dominated the field for a range of applications. The resolution of these models has subsequently increased from regional scales (typically 50 km) to kilometer and sub-kilometer scales. Online-coupled meteorology-chemistry frameworks, which emerged in the 2000s, further integrated gas-phase and aqueous chemical and dynamical processes[19,20].

Emission Inventories: From Static Data to Dynamic Inputs

In parallel with model development, emission inventories evolved from coarse national totals to sector-resolved, spatially gridded, and temporally varying datasets. Improvements in activity data, emission factors, and reporting requirements have greatly enhanced model realism. A modern prediction system requires inventories that include anthropogenic sources (transportation, industry, power generation, residential combustion, waste), emissions of volatile organic compounds (VOCs) from vegetation, emissions from biomass burning, increasingly derived from satellite fire detection, and natural sources such as dust and sea salt[21–23]. Bottom-up inventories are based on estimated activity data and emission factors, while top-down inventories are derived by inverting observed atmospheric concentrations through appropriate transport models[24].

Despite progress, such as the expansion of ground-based monitoring networks and the availability of satellite remote sensing information, uncertainties in emissions remain a dominant source of forecast error. This is particularly true in regions where activity information is limited and where episodic sources such as wildfires and agricultural burning are significant. Near real-time updating of emissions has therefore become an active frontier of operational development.

The Observational Revolution: In-Situ Networks and Satellite Remote Sensing

Observations are an important component of air quality forecasting. The expansion of ground-based regulatory networks during the late twentieth century provided high-accuracy, high-frequency measurements of surface pollutants such as $PM_{2.5}$, PM_{10}, ozone, nitrogen oxides, sulfur dioxide, and carbon monoxide, supporting both regulatory compliance and model evaluation[25]. These networks remain unevenly distributed globally, with dense coverage in high-income regions and sparse or incomplete coverage across large parts of the developing world. More recently, low-cost sensors, mobile monitoring platforms, and community-based observing networks have begun to complement regulatory networks by increasing the spatial density and flexibility of urban observations[26]. Although these systems generally do not match regulatory instruments in stability, calibration, or traceability, they provide information on street-level gradients, short-term episodes, and exposure-relevant variability that sparse fixed-site networks cannot resolve. Further, due to their relatively short life expectancy and in order to minimize errors that directly propagate in the AI models, these low-cost sensors need to be frequently replaced and sometimes recalibrated, which requires resources and infrastructure.

Satellite remote sensing has fundamentally transformed the landscape by providing systematic, global and near-real-time observations of atmospheric composition. Since the late 1990s, instruments measuring aerosol optical depth, tropospheric trace gases, and fire radiative power have transformed air quality science from a regional into a global enterprise, revealing transboundary transport, episodic plumes, and emission trends that surface networks alone could not resolve. Long-term satellite records have documented substantial NO_2 and SO_2 variations, directly informing policy evaluation[27]. The emergence of geostationary air quality satellites observing Asia (Geostationary Environment Monitoring Spectrometer or GEMS[28]), North America (Tropospheric Emissions: Monitoring of Pollution or TEMPO[29]), and Europe (Copernicus Sentinel-4[30]) at hourly resolution represents an important advance: diurnal cycles, rapid transport, and the temporal evolution of emission patterns can now be resolved from space. In short, satellite monitoring has become a core pillar of operational air quality services.

Data Assimilation to Improve Operational Forecasting

Borrowing techniques from numerical weather prediction, data assimilation methods (e.g., optimal interpolation, ensemble Kalman filters, and four-dimensional variational approaches) were introduced to combine models and observations in a statistically consistent manner[31,32]. By continuously adjusting modeled atmospheric composition towards observations, assimilation systems improve initial conditions, reduce systematic biases, and partially compensate for emission and transport errors. From an operational standpoint, data assimilation enables consistency across scales: global models supply boundary conditions for regional systems, which in turn drive urban-scale applications, with observational constraint propagating throughout the cascade. By the early 2000s, agencies in several countries had begun to operate daily air quality forecasts integrating CTMs, meteorological models, and observations.

Emergence of Artificial Intelligence

In the last five years, AI methods have become increasingly important components of some air quality forecasting systems[33]. They have been used for statistical post-processing and bias correction of CTM output, for emission estimation and inversion by combining satellite, surface, meteorological, and activity data, and for the construction of surrogate models that emulate computationally expensive CTMs at greatly reduced cost. AI methods, however, remain limited by their inability to extrapolate knowledge in unknown space in which the data for the system's training are not available. Hybrid configurations that combine physics-based models and machine learning components are therefore being increasingly promoted as the most robust way forward[34]. AI has not replaced physical understanding, but it has expanded predictive capabilities, particularly at urban scales and short lead times, and has begun to reshape the design of forecasting systems. Further, the AI approach, which is often advertised as drastically reducing computational costs, often relocates a non-negligible fraction of the resources in the training process and the treatment of large databases. At present, the operational use, validation, generalization and transferability of AI methods remain heterogeneous; progress is needed to bring this methodology to the required readiness level.

Integration

Incremental developments in operational air quality forecasting have outpaced the single model and empirical approaches of the 1950s and 1960s. However, the integration remains somewhat limited or superficial in important respects. The model components were developed at different times, by different communities, for different primary purposes, and the interfaces between them have rarely been the subject of dedicated scientific investment. The central question is therefore no longer simply how to improve each component, but how to design prediction systems in which components exchange information consistently across scales, disciplines, and decision contexts.

Challenges

Despite impressive progress over the past few decades, current air quality models face a number of important challenges that limit their ability to respond to societal needs. In many cases, the challenges facing urban air quality prediction are mismatches between the different model components, between scientific output and decision-relevant information. We organize these challenges into three categories dealing with model components, at the component interfaces, and forecasting objectives.

Component-Level Challenges

Significant uncertainties remain in every major component of the air quality prediction system. Emission inventories, for example, continue to misrepresent reality, sometimes by a factor 2–3, especially in urban areas where unreported sources (e.g., underreporting of traffic, construction dust, port and shipping plumes, and undeclared industrial waste, residential and other economic activities) escape inventory protocols[35]. The diurnal and weekly resolutions of assumed emissions are often poorly accounted for, which is reflected in the model projections. Emerging low-cost and mobile sensor networks partly address the spatial sparsity of conventional monitoring, but they introduce their own uncertainties related to calibration drift, cross-sensitivity, siting, representativeness, and data quality control, making effective forecasting dependent on rigorous calibration and integration with physically consistent models. Satellite observations, while providing quasi global coverage once or twice a day, do not provide the desired vertical resolution with large errors in the estimated species concentrations in the boundary layer[36,37]. Model representation of urban vegetation is often surprisingly primitive, typically reduced to drag coefficients and bulk deposition sinks, despite the recognized role of vegetation in emitting biogenic VOCs, modifying turbulence and humidity, trapping particulate matter, and modulating ozone formation[23,38].

Interface-Level Challenges

More structural, and less commonly discussed, are the limitations that arise at the temporal and spatial interfaces between model components. Multi-scale coupling remains a complex question. The downscaling from global models to mesoscale weather prediction to urban large-eddy simulation transfers information only in one way — from larger to smaller scales — with persistent inconsistencies in turbulence statistics across the nesting boundary, in aerosol and trace-gas composition, in temporal synchronization, and in boundary conditions[20,39,40]. Cities, which modify humidity, convection, cloud formation, precipitation, and heat island circulation, feed back to mesoscale meteorology in ways that current frameworks rarely allow. Street-scale chemistry presents a particularly clear interface failure. Current operational frameworks treat the chemistry of the urban canopy as if it were a downscaled regional chemistry, ignoring the structural reorganization that the urban environment imposes on photochemical pathways.

Another interface limitation is the incomplete integration of air-quality prediction with weather and climate services. Many high-impact urban episodes arise from the co-evolution of chemical pollution, meteorological extremes, and longer-term climate stressors, such as stagnation events, heat waves, wildfires, dust storms, and ozone episodes. Yet operational systems often treat air quality, weather hazards, and climate adaptation as separate service streams, with different models, temporal and spatial scales, uncertainty frameworks, and decision interfaces. This separation limits the ability of forecasts to support compound-risk assessment and coordinated urban response.

The interface between machine learning and physical models is an emerging challenge that requires attention[33]. The dominant pattern in implementation is loose coupling, where machine learning post-processes output from a physical model, replaces a particular module or interpolates between observations. Commonly used skilled interpolators such as random forests, gradient boosting machines, and recurrent neural networks do not account for conservation laws, reaction kinetics, or turbulent transport[41,42]. When the underlying physical or chemical regime (e.g., due to climate change, the implementation of emission controls, or unusual weather patterns) shifts, these models extrapolate without constraints. The integration of data-driven and mechanistic components rarely includes the introduction of physical constraints into the training process itself.

Objective-Level Challenges

A third and deeper question concerns what the prediction system is predicting. Current systems predict pollutant concentrations on static grid cells, while people move through the city, transitioning between indoor and outdoor environments at different breathing rates, with susceptibility that depends on age, health, occupation, and concurrent exposures such as heat[43–45]. The relevant quantity for health effects is not the concentration of pollutants or allergens at a fixed location, but the accumulated dose along an individual trajectory[46]. The gap between modeled concentration and accumulated personal dose, for example, is large and systematic, and cannot be improved by finer spatial resolution alone.

The representation of uncertainty is another issue. Most operational outputs convey a precision in predictions that is not supported by the underlying uncertainties in emissions, chemistry, turbulence, meteorology, and observations[47,48]. Operational systems should therefore communicate to the public a degree of probability that a relevant threshold will be exceeded over a specified period.

Finally, it is important to remember that the forecasting skill is highest in regions where observation networks are dense and institutional investment is sustained. Regions with the highest exposure to pollution tend to be those with the least established forecasting capabilities[49,50]. In addition, AI components are trained primarily on data from instrumented regions and may widen rather than narrow this gap between regions.

These three categories of challenges — component level, interface level, and goal level — are not independent. Component constraints propagate through interfaces, and interface constraints distort the goals that the system can credibly address. The directions outlined in the next section suggest that progress at any one level depends on integration with the others.

New Directions: Towards Intelligent Analysis and Prediction Systems

The challenges identified above point to a common conclusion: the next stage of urban air quality prediction will not only depend on refining individual components but also on redesigning the interfaces among models, observations, human activity, infrastructure, decision systems, and broader weather–climate–environment services.

With this objective in mind, we identify ten emerging research directions that, taken together, are reshaping urban air quality prediction towards an integrated and intelligent approach. Such a system should also provide information needed to analyze the processes that are responsible for the past and predicted changes in the urban system, and to evaluate the potential consequences of planned measures taken by decision-makers at all governance levels. These directions are not isolated technical advances; each addresses one or more of the interface failures discussed above, and several are co-dependent, and together they define the integrated urban digital twin framework illustrated in Fig. 2. We organize them into four groups according to the type of interface they target: human–nature interfaces, physics–AI interfaces, observation–model interfaces, and scale and system interfaces.

Human–Nature Interfaces

Direction 1: From concentration maps to exposure fields. A radical reframing of the prediction goal is the transition from concentration fields to dynamic personal exposure fields. If focusing only on outdoor exposure rather than accounting also for the larger indoor pollution effects, the coupling of urban computational fluid dynamics with mobility data, transport network information, indoor-outdoor exchange models, and physiological variables must be simultaneously taken into account. Such systems shift the policy-relevant question from "What is the level of pollution in the city?" to "Who is getting the highest dose, when, and why?". Real-time exposure twins could support targeted protection of vulnerable groups such as school children, outdoor workers, and the elderly, and are conceptually aligned with the broader shift towards impact-based forecasting. However, to ensure privacy of individuals and limit data governance difficulties associated with population-scale tracking, the forecasting infrastructure should preferably use synthetic information provided by activity-based models and surveys rather than data generated by personal sensors, mobile phones or GPS tracing systems.

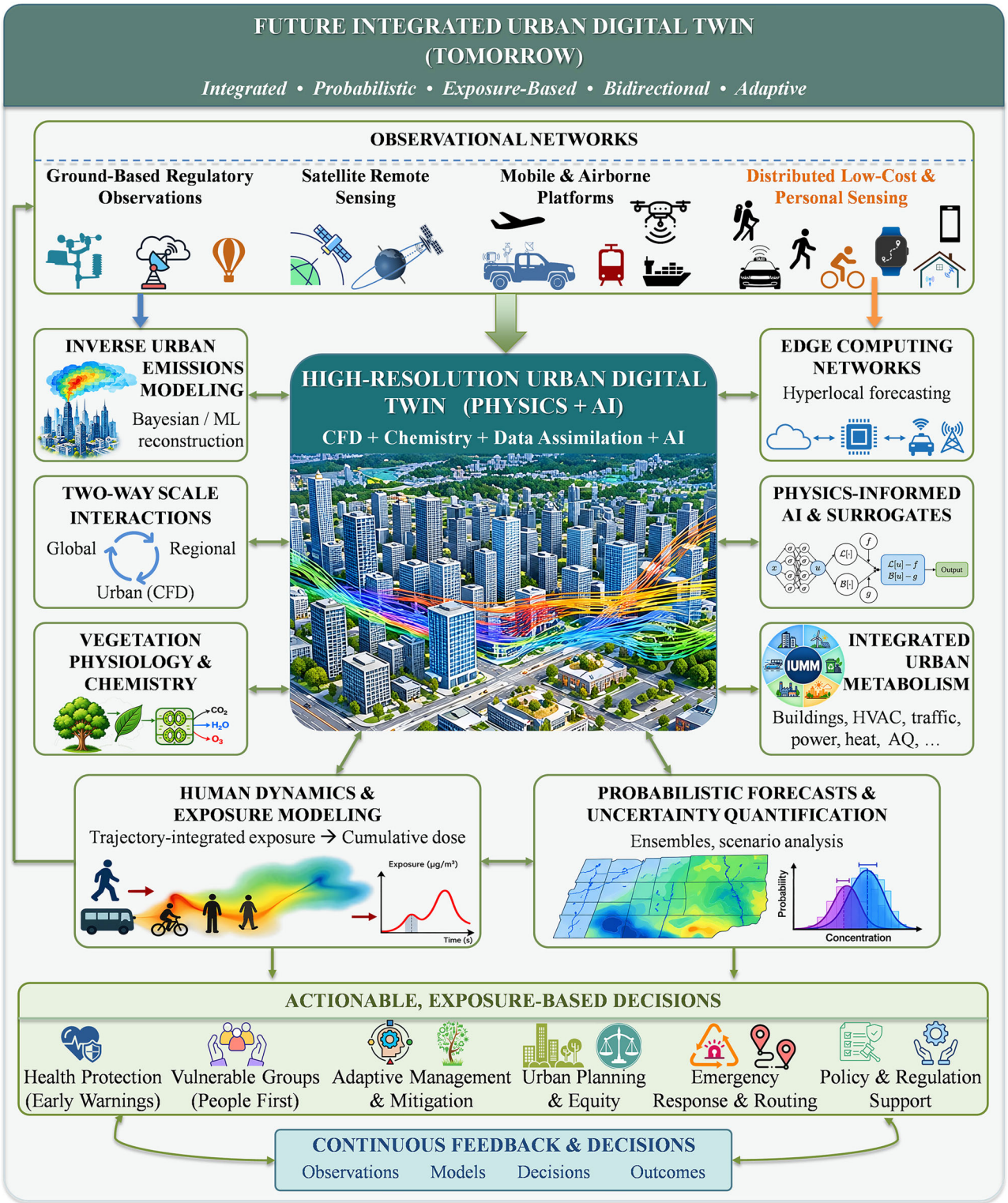


Fig. 2 | Future integrated urban digital twin framework. The framework places a high-resolution urban digital twin at the core, coupling computational fluid dynamics (CFD), atmospheric chemistry, data assimilation and artificial intelligence (AI). Observational networks continuously constrain and update the central system. Around this core, inverse emission estimation, two-way scale coupling, vegetation–physiology–chemistry interactions, physics-informed AI surrogates, edge computing, urban metabolism, and human dynamics and exposure modelling provide complementary information and processes. These coupled components generate probabilistic forecasts, uncertainty estimates and exposure information to support health protection, adaptive management, urban planning, emergency response, and policy and regulation. Feedback among observations, models, decisions and outcomes enables continuous updating and adaptation of the prediction system.

Direction 2: Agent-based pollution systems and human–atmosphere feedbacks. Human activity is not an exogenous forcing on urban atmospheric chemistry, but part of a coupled and adaptive system. Automobile drivers reroute in response to congestion; pollution warnings change behavior; and these behavioral responses feed back into emission and exposure patterns. Such loops produce emergent pollution dynamics that static inventory-based models cannot capture. Agent-based representations of urban mobility, coupled with reinforcement learning of human response and CFD surrogate models of pollutant dispersion, allow these feedbacks to be represented dynamically rather than prescribed as fixed emission scenarios[51,52]. This direction represents the most direct route to closing the loop between the atmospheric and behavioral components of the urban environment.

Physics–AI Interfaces

Direction 3: Artificial intelligence that respects turbulence physics. Urban turbulence is highly intermittent and anisotropic, especially within building canopies, vegetated zones, road intersections, and buoyancy-dominated summertime flows, where conventional Reynolds-averaged closures often fail. Fully resolving the relevant eddies remains too expensive for operational prediction. This creates a natural role for AI, not as a substitute for physical modelling, but as a means of learning closures, emulators, and reduced-order operators from high-fidelity simulations and observations. Promising approaches include neural closures for large-eddy simulation (LES), machine-learned subgrid-scale parameterizations, differentiable Navier-Stokes solvers, and operator-learning frameworks that approximate the action of partial differential equations[53,54]. Deriving unresolved turbulent transport from LES output, while constraining this derivation by conservation laws and asymptotic bounds, could reduce the computational time of city-scale CFD from days to minutes while remaining physically interpretable — a combination that neither purely mechanistic models nor purely data-driven approaches have achieved alone. Such models could enable near real-time digital twins of the urban environment.

Direction 4: Hyperlocal forecasting with edge computing. Centralized supercomputer model simulations are increasingly being augmented by distributed sensing and artificial intelligence. Sensors mounted on buses, drones, transportation infrastructure, smartphones, and building rooftops can feed localized LES surrogates, graph neural network nowcasts, and adaptive turbulence corrections to produce block-by-block predictive air quality fields updated on minute timescales[26,55]. The technical infrastructure (edge processors, low-power neural network inference, mesh networking) is feasible; the scientific challenge is to design model architectures that distinguish signal from local artifact and maintain consistency with regional forecasts, emission context, and boundary conditions. The surrogate models that enable hyperlocal forecasting must themselves be physically informed (direction 3) if they are to remain reliable under non-stationary conditions.

Observation–Model Interfaces

Direction 5: Inverse urban emissions from multi-source observations. Because urban emissions remain among the most uncertain inputs in forecasting systems, a major direction is to treat them as dynamically inferred quantities rather than fixed boundary conditions. Instead of imposing emissions and predicting concentrations, inverse approaches use observations to infer emissions consistent with what is measured[56]. Modern formulations combine low-cost sensor networks, satellite retrievals, lidar profiles, traffic camera streams, meteorological reanalyzes, and turbulence models within Bayesian source reconstruction frameworks. The recent integration of urban CFD with graph neural networks and differentiable physics has dramatically expanded the feasibility of this approach for complex urban geometries[34,57]. Applications include, for example, the determination of illegal industrial emissions, the quantification of underreported traffic activity, the resolution of port and shipping plumes, the attribution of construction-related dust, and other localized, episodic, or poorly inventoried urban sources, including some waste-management activities[58]. In this sense, the city's sensor network becomes a distributed monitoring system for its own emissions behavior.

Direction 6: Uncertainty-aware urban prediction and probabilistic modelling. A central requirement for next-generation urban prediction systems is to produce probabilistic outputs in which uncertainty is identified, attributed, and propagated across the full modelling chain[59,60]. Emission estimation, sensor calibration, meteorological forcing, turbulent transport, chemical mechanisms, boundary conditions, and exposure-related activity data each carry uncertainties that may amplify or compensate for one another as they pass through coupled models. The goal is not only to produce probabilistic concentration maps, but to understand how uncertainty enters the system, how it is transformed at component interfaces, and how it should be communicated to decision makers. The technical components (e.g. ensemble LES, Bayesian inference, stochastic turbulence parameterization, structured uncertainty propagation) are well established in adjacent fields, but computationally expensive at urban scales. Their adoption depends critically on the development of fast surrogate models (Direction 3) and the willingness of operational agencies to communicate probabilistic information in actionable form.

Scale and System Interfaces

Direction 7: Street canyons as reactive chemical reactors. Treating dense urban streets as passive traffic corridors obscures a significant amount of relevant chemistry. Street canyons trap solar radiation, accumulate NO_X and VOCs emitted at street level, maintain intermittent turbulent mixing, and host, for example, $NO–NO_2–O_3$ cycling under photochemical regimes that differ from the regional background[61]. Recent advances in GPU-accelerated chemistry solvers, operator splitting, and machine-learned chemical surrogates make block- and facade-scale chemistry coupled with LES computationally tractable[62,63]. In a new approach, a street canyon can be treated as a node in a micro-reactor network (with urban morphology acting as a chemistry controller), analogous to reactor geometry in combustion engineering. In such a design, canyon aspect ratio, building height variability, and street orientation would modulate not only dispersion but also chemistry.

Direction 8: Two-way meso–microscale coupling. Cities modify heat, moisture, momentum, and chemistry in ways that are increasingly recognized as climatologically and chemically significant. True two-way meso-microscale coupling, in which urban modifications are propagated upward into mesoscale evolution, requires architectural changes in modelling chains, careful treatment of turbulence statistics across the nesting boundary, and consistent aerosol and trace gas composition across the coupling interface[4,40,64]. The methodological requirements for bidirectional coupling are demanding, but the alternative is a persistent mismatch between the scales at which cities are modeled and the scales at which urban climate evolves.

Direction 9: Vegetation–turbulence–chemistry coupling. Urban vegetation modulates the atmosphere through multiple coupled pathways. Trees and other vegetation emit biogenic VOCs and particles such as pollen, alter turbulence statistics, modify humidity and thermal stratification, deposit particulate matter, and influence street ventilation[65,66]. The net effect on air quality is non-trivial: vegetation can reduce particulate concentrations while increasing ozone precursors, improve thermal comfort while reducing street ventilation, and stabilize the local boundary layer in a way that traps pollutants. Embedding physiological vegetation models (i.e., stomatal conductance, seasonal phenology, drought response, species-specific biogenic VOC emission factors) directly into urban LES is therefore critical to the credibility of green infrastructure policy.

Direction 10: Integrated urban metabolism: air quality, energy, and heat. Future cities increasingly operate as integrated urban metabolism–atmosphere systems, in which energy use, material flows, waste processing, mobility, and infrastructure operation jointly shape emissions, urban heat, atmospheric stability, and air quality. This coupling will become more pronounced as cities adopt electrified transportation, distributed heat pumps, district heating networks, building heating, ventilation and air conditioning (HVAC) systems, waste heat reuse from data centers, and emerging hydrogen infrastructure. Most current air quality models treat these as static emission inputs or external boundary conditions, missing the essential coupling between them. Integrated urban metabolism modelling reimagines the city as a coupled fluid-energy-chemical system[67]. This broad direction provides a framework within which questions about decarbonization pathways, co-benefits of electrification, and climate adaptation can be answered with chemical and dynamical consistency.

These directions differ in maturity — some, such as exposure mapping and emission inversions, are already producing peer-reviewed demonstrations; others, such as integrated urban metabolism, remain conceptual, but together they point towards a more integrated prediction architecture.

The Way Forward

Urban air quality forecasting has evolved from empirical detection of pollution episodes to integrated systems that combine mechanistic models, comprehensive surface and space observations, data assimilation, and, increasingly, artificial intelligence. While this trajectory has been successful, it should be recognized that further refinement along the established directions — more chemistry, finer grids, more parameters — will yield only marginal returns. In this Perspective, we suggest that future progress will result from a different kind of investment: integration at the interfaces between components, between disciplines, and between forecasts and the human and infrastructure systems they are intended to serve.

The directions outlined are interdependent: physics-informed AI enables fast surrogate models for urban CFD, making it feasible to generate probabilistic concentration and exposure maps; explicit uncertainty quantification, in turn, gives operational credibility to top-down emission inversions. Bidirectional meso-microscale coupling provides the consistent boundary conditions needed for street-canyon chemistry, and vegetation-turbulence coupling and urban metabolism jointly determine whether urban climate adaptation strategies improve or degrade air quality. These connections suggest that the ten directions should not be pursued as separate research tracks, but as mutually reinforcing components of a coherent prediction architecture.

For such systems to become operationally meaningful, their development must involve closer interactions with city managers, public-health agencies, infrastructure operators, and communities, so that forecasts of concentration, exposure, uncertainty, and system state can be translated into feasible interventions, risk communication, and evaluation of policy trade-offs. The decision interface is therefore not an afterthought to prediction, but part of the prediction system itself.

Of particular concern is the global geography of capabilities. Forecasting expertise, computational resources, observation density, and the data on which machine learning components are trained remain concentrated in regions where pollution exposure is not the highest. The transition to integrated and intelligent urban forecasting systems must not exacerbate this asymmetry. Open frameworks, shared model infrastructure, internationally coordinated observational standards, and the development of methods that work reliably under data-sparse conditions are technical contributions to global equity, not separate policy goals.

The maturation of urban air quality prediction will not be defined by model complexity alone, but by the integration of these models with each other, with the observations that constrain them, with the human activities they describe, and with the decisions they are intended to inform. The transition from air quality forecasting to integrated and intelligent urban forecasting systems is ultimately a reconfiguration of how the field understands its purpose. Such an approach will enable societies to meet energy demands and sustainable development needs while protecting human health and climate, and in short, to improve living conditions in all parts of the world.

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Author contributions

G.B. initiated the conceptual framework. Y.W. prepared the figures. Y.W. and G.B. jointly wrote the manuscript. Both authors contributed to manuscript revision and approved the final version.

Competing interests

The authors declare no competing interests.

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